

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

Compiled by : *R. Nuga*

Q1:JUN 2008 P2

6 (a) Define the term

(i) *capacitance,*

(ii) *the farad.*

[2]

(b) A student is given two $4\ \mu\text{F}$ capacitors. Draw diagrams to show how the capacitors could be connected in order to give

(i) $8\ \mu\text{F}$ capacitance,

(ii) $2\ \mu\text{F}$ capacitance.

[2]

(c) A capacitor of capacitance C is fully charged by a $200\ \text{V}$ battery. It is then discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity $2.5 \times 10^{-2}\ \text{Jkg}^{-1}\text{K}^{-1}$ and mass of $0.1\ \text{kg}$.

If the temperature of the block rises by $0.4\ \text{K}$, find the value of C .

$C =$ _____ [3]

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Q2:JUN 2010 P2

5 (a) Define

1. *capacitance,*

2. *the farad.*

[2]

(b) A capacitor is marked “220 μ F; 20 V”.

(i) Explain the importance of the 20 V.

(ii) Calculate the

1. charge stored by the capacitor,

charge = _____

2. energy stored by the capacitor.

energy = _____

[5]

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

Compiled by : *R. Nuga*

Q3:NOV 2011 P2

6 (a) Define a capacitor.

[1]

(b) A 6 V battery is used to charge a $4\ \mu\text{F}$ capacitor.

(i) Calculate

1. the energy supplied by the battery,

2. the energy stored by the capacitor.

(ii) Comment on the differences between your answers to (b)(i).

[5]

Q4:NOV 2014 P2

6 (a) Define

(i) capacitance,

(ii) the farad.

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

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- (b) If a student was given two $4\ \mu\text{F}$ capacitors, draw diagrams to show how the capacitors could be connected in order to give a combined capacitance of

(i) $2\ \mu\text{F}$,

(ii) $8\ \mu\text{F}$.

[2]

- (c) A capacitor of capacitance, C , is fully charged by a 200 V battery. It is then discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity $2.5 \times 10^2\ \text{Jkg}^{-1}\text{K}^{-1}$ and mass 0.1 kg.

If the temperature of the block rises by 0.4 K, find the value of C .

$C =$ _____ [3]

Q5:JUN 2016 P2

- 5 (a) Define *the farad*.

[1]

- (b) Fig. 5.1 shows a $15\ \mu\text{F}$ capacitor connected in series with $1600\ \Omega$ resistor and $20\ \text{V}$ supply of negligible internal resistance.

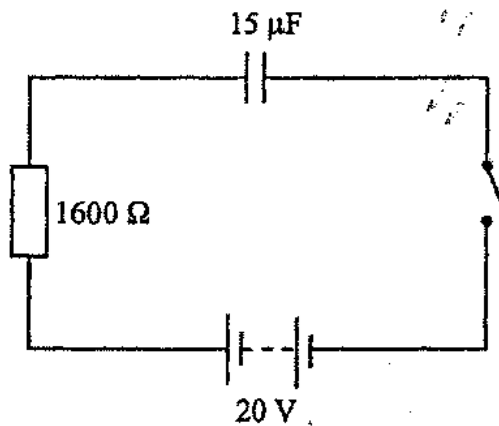


Fig. 5.1

The capacitor is initially uncharged.

Calculate the

- (i) current in the circuit immediately after the switch is closed,

current = _____

- (ii) final charge on the capacitor.

charge = _____

[3]

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

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- (c) The capacitor in (b) is then connected in parallel with an uncharged $18.0 \mu\text{F}$ capacitor.

Calculate the p.d across the $18 \mu\text{F}$ capacitor.

p.d = _____ [2]

Q6:NOV 2018 P2

- 6 Fig. 6.1 shows a circuit used to charge a capacitor using a 6.0 V battery of negligible internal resistance. The capacitor is initially uncharged.

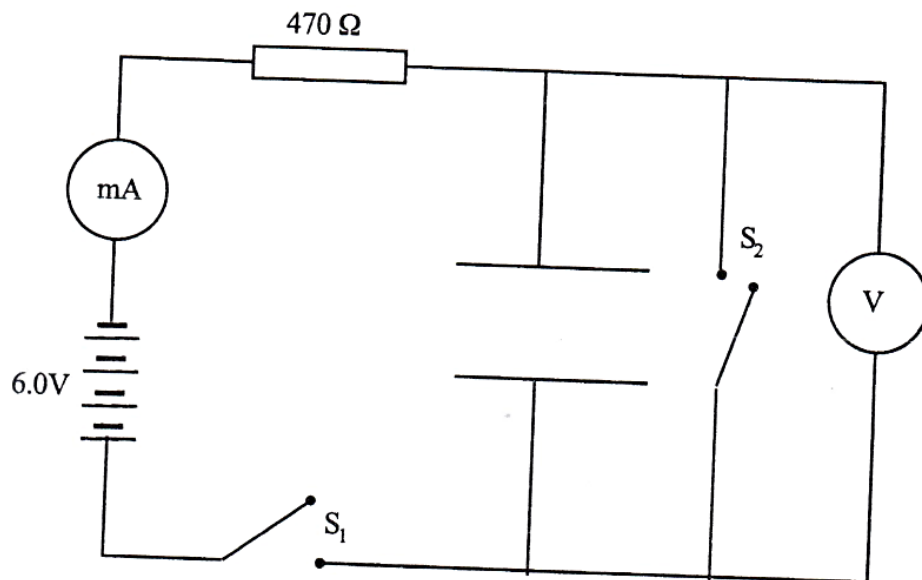


Fig. 6.1

- (a) If S_2 is opened and S_1 is closed, sketch a graph to show the variation with time of the
- (i) voltmeter reading,
 - (ii) ammeter reading.

[2]

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

Compiled by : *R. Nuzo*

- (b) (i) Calculate the ammeter reading on closing S_1 with S_2 open.

- (ii) State the voltmeter and ammeter readings when the capacitor is fully charged.

Voltmeter reading _____

Ammeter reading _____ [4]

- (c) Suggest the purpose of switch S_2 .

_____ [1]

Q7:JUN 2008 P3

- 4 (a) Define *capacitance*. [1]

- (b) Capacitors have been recently developed which have capacitances of 3.3 F and dimensions 43 mm × 43 mm × 15 mm. They are used for maintaining current in computer memories when the computer is switched off.

- (i) Calculate the energy per unit volume of the capacitor when it is charged up to 5.0 V.

- (ii) Compare your answer in (i) to the energy per unit volume of petrol. (heat of combustion = $4.8 \times 10^7 \text{ J kg}^{-1}$ and density = 700 kg m^{-3})

- (iii) A capacitor of capacitance, C , supplies a current, I , for a time, t .

Show that the drop in the voltage, ΔV , is given by $\Delta V = \frac{I\Delta t}{C}$.

- (iv) The leakage current of the capacitor is 0.40 mA.

Determine the time taken for the p.d. across the capacitor to drop from 5.0 V to 3.5 V.

- (v) Explain how you would use another 3.3 F capacitor to produce twice the time in (iv).

[13]

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- (c) Fig. 4.1 shows a connection of two capacitors.

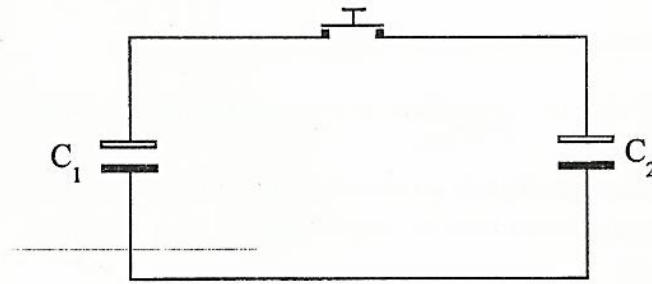


Fig. 4.1

C_1 is fully charged and C_2 is not charged.

Explain what happens to each of the following when the switch is closed.

- (i) total charge
- (ii) potential difference
- (iii) total energy stored

[6]

Q8:NOV 2006 P3

- 4 (a) Define *capacitance* and state its SI unit. [2]
- (b) (i) Show that energy gained by a capacitor during charging is given by $W = \frac{1}{2}CV^2$, where C is capacitance and V is potential difference.
- (ii) Calculate the total capacitance of the circuit in Fig. 4.1 if each capacitor has a capacitance of $3.0 \mu\text{F}$.

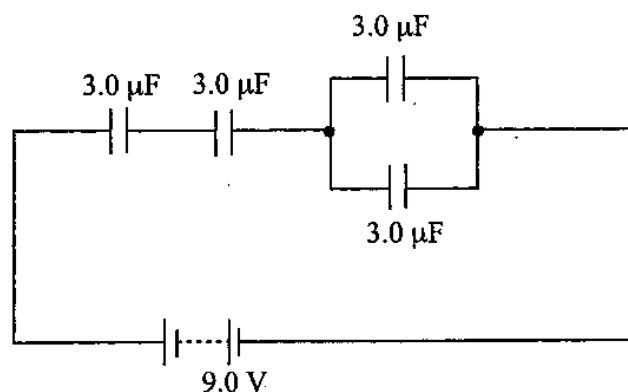


Fig 4.1

- (iii) A 9.0V battery is used to fully charge the capacitor. Calculate the total energy gained by the capacitors.

[7]

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

Compiled by : *R. Muga*

Q9:NOV 2010 P5

- 5 (a) Define (i) *capacitance*,
(ii) *electric potential*.

[2]

- (b) An isolated metal sphere of radius, r , has a charge, Q , concentrated at its centre.
(i) Give an expression of the electric potential, V , at the surface of the sphere.
(ii) Hence show that the capacitance, C , of the capacitor is

$$C = 4\pi\epsilon_0 r.$$

[3]

- (c) A capacitor marked " $15\mu\text{F}$, 20 V " is charged by connecting it to a 3 V battery.
(i) Explain the significance of 20 V .
(ii) Calculate the
1. energy supplied to the capacitor,
2. energy stored by the capacitor.
(iii) Comment on your answers in (ii) above.

[7]

Q10:JUN 2016 P5

- 5 (a) (i) Define *capacitance*.
(ii) A capacitor has a charge, Q , and a potential difference, V .

Show, with the aid of a diagram, that the energy, W , stored by the capacitor is given by the formula:

$$W = \frac{1}{2}QV$$

[4]

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

Compiled by : R. Naga

- (b) Fig. 5.1 shows an electric circuit.

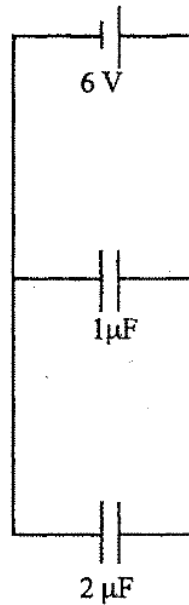


Fig. 5.1

Determine the

- (i) potential difference across each capacitor.
- (ii) total charge stored.
- (iii) energy stored in the $1 \mu\text{F}$ capacitor.

[6]

- (c) Give any two applications of capacitors in industry.

[2]

Q11:JUN 2015 P5

- 2 (a) Define the farad. [1]
- (b) Two identical capacitors are to be connected separately to form a series connection and a parallel connection.
- (i) Derive an expression for the effective capacitance in each case.
- (ii) State the laws from which the derivations are based.

[6]

(c) Fig.2.1 shows a combination of capacitors in a circuit.

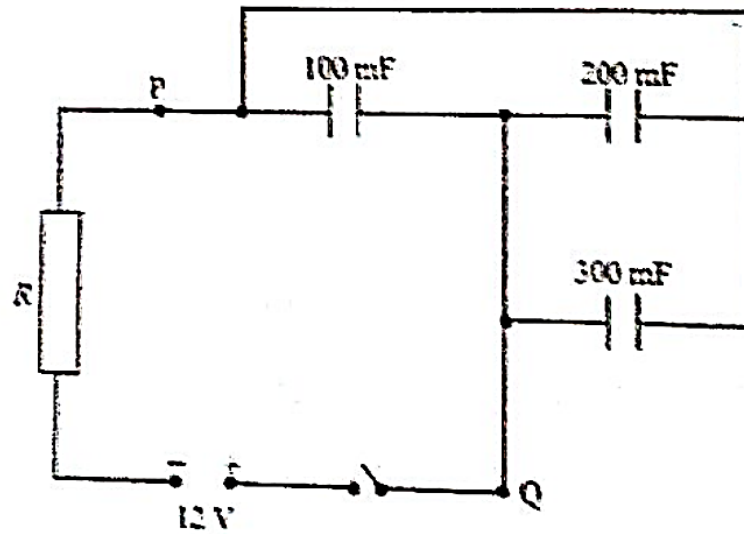


Fig 2.1

(i) Determine

1. the effective capacitance between points P and Q.
2. the total energy stored when fully charged.

(ii) In Fig. 2.1 the switch is closed and the time taken to fully charge the capacitor is noted.

State the effect of increasing the resistance, R , on the time taken to fully charge the capacitors.

[4]

(d) On most electrical gadgets there is a warning sign not to open the gadget unless you are qualified to do so.

Suggest an explanation for this warning.

[2]

Q12:NOV 2015 P5

- 1 (c) (i) Define *capacitance*.
- (ii) A capacitor is marked ' $100 \mu\text{F}$, 12 V '.

Calculate the

1. energy,
2. charge,

stored by the capacitor when a p.d. of 12 V is applied across it.

[5]

Q13:JUN 2014 P5

- 1 (b) The circuit in Fig. 1.2 shows a charged capacitor connected in series with a resistor of resistance, R , and a switch S .

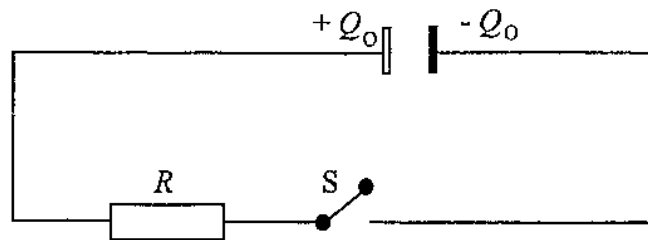


Fig. 1.2

- (i) Sketch a graph of charge against time from the instant the switch is closed.
- (ii) Show, on the same axes, the effect of
 1. increasing capacitance and label it 1.
 2. using thicker connecting wires and label it 2.
- (iii) Explain the effect in (ii) 2.

[4]

Q14:JUN 2010 P5

- 1 (b) (i) Use the definitions of electric current and capacitance to show that
- $$C = \frac{I}{fV},$$
- where f is frequency and other symbols have their usual meanings.
- (ii) Fig. 1.1 shows a circuit used to investigate the properties of a capacitor. The capacitor is initially not charged.

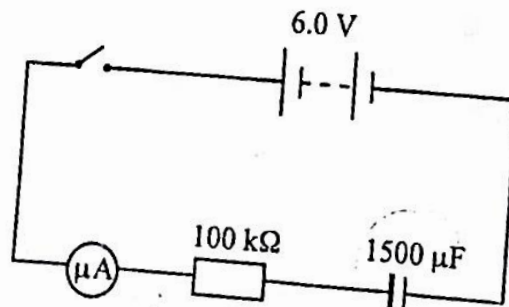


Fig. 1.1

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When the switch was closed, the pointer of the ammeter was seen to deflect but after 4.0 minutes there was no deflection at all.

1. Explain this observation.
2. Sketch a graph to show how the voltage, V , across the $100\text{ k}\Omega$ resistor varies with time.
3. Calculate the value of the current which flows in the circuit at the instant of closing the switch.

Q15:JUN 2018 P5

4 (a) Define the terms

1. *capacitance*,
2. *farad*.

[2]

(b) (i) Show that for capacitors C_1 , C_2 and C_3 connected in series,

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

(ii) Fig. 4.1 shows three capacitors connected.

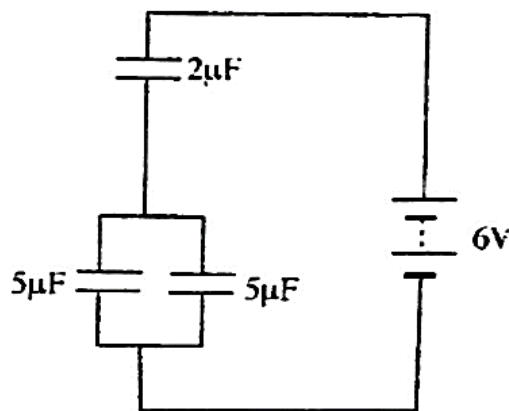


Fig. 4.1

4.4 CAPACITANCES P2, P3&P5 QUESTIONS

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Calculate the

- 1. charge of each capacitor,**
 - 2. energy stored by the $5\ \mu\text{F}$ capacitor. [8]**
- (c) Explain why the energy supplied to a capacitor is always more than the energy stored by the capacitor. [2]**

Q16:JUN 2017 P5

- (c)**
- (i)** Define *capacitance* and *the farad*.
 - (ii)** Hence derive the formula for combined capacitance C_T for three capacitors of capacitances C_1 , C_2 and C_3 connected in parallel.
 - (iii)** Draw a diagram to show how three capacitors, each of capacitance $4.0\ \mu\text{F}$ can be connected to get a combined capacitance of $6.0\ \mu\text{F}$.

[6]